# Ho-Joong Kim

Ph.D. Candidate

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## Education

**Korea University** *Mar. 2021 - PRESENT*

Ph.D. in Artificial Intelligence

Advisor: Prof. Seong-Whan Lee

**Hansung University** *Mar. 2015 - Feb. 2021*

B.S. in Industrial Management Engineering and Computer Engineering

## Publications

**TF-CADE: Foreground-Concentrated Text-Video Alignment for Zero-Shot Temporal Action Detection** ***CVPR**, 2026*

Yearang Lee, **Ho-Joong Kim**, and Seong-Whan Lee *Highlight*

**DiGIT: Multi-Dilated Gated Encoder and Central-Adjacent Region Integrated Decoder for Temporal Action Detection Transformer** ***CVPR**, 2025*

**Ho-Joong Kim**, Yearang Lee, Jung-Ho Hong, and Seong-Whan Lee

**Comprehensive Information Bottleneck for Unveiling Universal Attribution to Interpret Vision Transformers** ***CVPR**, 2025*

Jung-Ho Hong, **Ho-Joong Kim**, Kyu-Sung Jeon, and Seong-Whan Lee *Highlight*

**MIRe: Enhancing Multimodal Queries Representation via Fusion-Free Modality Interaction for Multimodal Retrieval** ***ACL**, 2025*

Yeong-Joon Ju, **Ho-Joong Kim**, and Seong-Whan Lee *Findings*

**TE-TAD: Towards Full End-to-End Temporal Action Detection via Time-Aligned Coordinate Expression** ***CVPR**, 2024*

**Ho-Joong Kim**, Jung-Ho Hong, Heejo Kong, and Seong-Whan Lee

**Text-Infused Attention and Foreground-Aware Modeling for Zero-Shot Temporal Action Detection** ***NeurIPS**, 2024*

Yearang Lee, **Ho-Joong Kim**, and Seong-Whan Lee

**Unknown-Aware Graph Regularization for Robust Semi-supervised Learning from Uncurated Data** ***AAAI**, 2024*

Heejo Kong, Suneung Kim, **Ho-Joong Kim**, and Seong-Whan Lee

**DCR-TSF: Dual-Level Compositional Reasoning with Triadic Spherical Fusion for Training-Free Zero-Shot Composed Image Retrieval** *ACCV, 2026*

Gwang-Ho Na, **Ho-Joong Kim**, and Seong-Whan Lee

**ProCal: Inference-Time Proposal Calibration for Open-Vocabulary Object Detection** *SMC, 2026*

Jae-Ryung Hong, **Ho-Joong Kim**, and Seong-Whan Lee

**DiCE-CIR: Direct Composition Learning for Efficient Zero-Shot Composed Image Retrieval** *SMC, 2026*

Gwang-Ho Na, **Ho-Joong Kim**, and Seong-Whan Lee

**ClipTBP: Clip-Pair based Temporal Boundary Prediction with Boundary-Aware Learning for Moment Retrieval** *ICPR, 2026*

Ji-Hyeon Kim, **Ho-Joong Kim**, and Seong-Whan Lee

**InterHier: Learning Interconnected Hierarchical Semantics for Open-Vocabulary Object Detection** *IEEE Access, 2026*

Yeong-Jin Kim, **Ho-Joong Kim**, and Seong-Whan Lee

**FIQ: Fundamental Question Generation with the Integration of Question Embeddings for Video Question Answering** *SMC, 2025*

Juyoung Oh, **Ho-Joong Kim**, and Seong-Whan Lee

**Ensuring Spatial Scalability with Temporal-Wise Spatial Attentive Pooling for Temporal Action Detection** *Neural Networks, 2024*

**Ho-Joong Kim**, and Seong-Whan Lee

**Description Attribute-Enhanced Spatio-Temporal Zero-shot Action Recognition** *ICPRAI, 2024*

Yehna Kim, **Ho-Joong Kim**, and Seong-Whan Lee

**Enhancing Discriminative Ability among Similar Classes with Guidance of Text-Image Correlation for Unsupervised Domain Adaptation** *IJCNN, 2023*

Yu-Won Lee, Myeong-Seok Oh, **Ho-Joong Kim**, and Seong-Whan Lee *Oral*

**Temporal-Invariant Video Representation Learning with Dynamic Temporal Resolutions** *AVSS, 2022*

Seong-Yun Jeong, **Ho-Joong Kim**, Myeong-Seok Oh, Gun-Hee Lee, and Seong-Whan Lee *Oral*

**SCGN: Novel Generative Model using the Convergence of Latent Space by Training** *Electronics Letters, 2020*

**Ho-Joong Kim**, and Sung-Hoon Jung

**SOGN: Novel Generative Model using Self Organizing Map** *Electronics Letters, 2019*

**Ho-Joong Kim**, and Sung-Hoon Jung

## Projects

**Developing Document Understanding VLLM Agent** *Aug. 2025 – Aug. 2026*

Konan Technology

Developing a **document layout analysis** model and **integrating VLLM** to understand analyzed document components. Designing a **lightweight neural network architecture** based on the developed model.

**Developing General-Purpose AI Model for Industrial Safety** *Sep. 2024 – Aug. 2025*

Mithril

Developed an **open-vocabulary object detection** model capable of detecting unseen objects using linguistic information. Designed a **lightweight neural network architecture** based on the developed model.

### Video Event Detection and Recognition via Unsupervised Learning *Mar. 2021 – Nov. 2023*

IITP

Designed an **self-supervised pre-training** model and **zero-shot action recognition** for video understanding using limited computing resources in an unsupervised learning setting.

### Lightweight Deep Neural Networks for Mobile Edge Computing *Jun. 2019 – Mar. 2020*

Prizzmable

Developed a model for **tooth object detection** and **plaque segmentation**. Designed **lightweight networks** optimized for deployment on **mobile and embedded devices**.

## Awards

### Best Paper Award *2019*

The Institute of Electronics and Information Engineers Summer Conference

### Best Paper Award *2018*

The Institute of Electronics and Information Engineers Fall Conference

## Patent

### Method and system for learning self-converging generative networks

*Republic of Korea, 2023*

*Patent No. KR102580159B1*